

Input parameters

Parameter	Value	Unit	Source
Positive electrode thickness	60.0	um	design override
Negative electrode thickness	68.0	um	design override
Separator thickness	8.0	um	design override
Pos CC (Al)	10.0	um	design override
Neg CC (Cu)	6.0	um	design override
Electrolyte conductivity	2.0	S/m	design override
Electrolyte diffusivity	5.0e-10	m2/s	design override
Cation transference number	0.60	-	design override
Heat transfer coefficient	40	W/m2K	design override
Nominal capacity	3.93	Ah	design override (true 1C)
Electrode area	0.1027	m2	0.065 x 1.58 m

Capacity and energy

Quantity	Value	Unit	Source
1C discharge capacity	3.9547	Ah	final_b7_1c_dfn.json (DFN)
Discharge energy	14.0377	Wh	$\int V \cdot I \, dt$
Midpoint voltage	3.8142	V	final_b7_energy_dfn.json
DC resistance	3.8541	mOhm	final_b7_energy_dfn.json
Low-T retention	0.99286	-	final_b7_retention_dfn.json

Energy density

Quantity	Value	Unit	Formula
Gravimetric ED	460.06	Wh/kg	energy / mass = 14.0377 / 0.030513
Volumetric ED	899.25	Wh/L	energy / volume(L) = 14.0377 / 0.015610
Requirement ED_kg	327.18	Wh/kg	entry 0 (min)
Requirement ED_L	880.0	Wh/L	entry 0 (min)

N/P and mass

Quantity	Value	Unit	Formula
N/P ratio	0.6711	-	$(c_{\text{max,neg}} \times \text{eps} \times t) / (c_{\text{max,pos}} \times \text{eps} \times t)$
Total mass (excl. electrolyte)	30.5126	g	Σ layers
Electrolyte mass	5.0356	g	pore vol x 1.2 g/cm3 (lit.)
Total incl. electrolyte	35.5482	g	formula caliber

Process parameters

Parameter	Value	Unit	Formula
Positive areal density	130.2	g/m2	$t \times (1-\text{por}) \times \rho$
Negative areal density	84.5	g/m2	$t \times (1-\text{por}) \times \rho$

Positive compaction density	2.17	g/cm ³	$\rho \times (1-\text{por}) / 1000$
Negative compaction density	1.24	g/cm ³	$\rho \times (1-\text{por}) / 1000$
Electrolyte fill amount	5.04	g/cell	pore vol x 1.2 g/cm ³ x fill 1.0